# STAFF REVIEW · H446 031 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

## Do Now — source answers

1. Sum = 1 because 1 XOR 0 = 1. Carry = 0 because 1 AND 0 = 0.

2. ¬A ∧ ¬B. Negate both inputs and change OR to AND.

3. 30 is the right child of 20: 30 is less than 40, but greater than 20.

4. The exponents already match. $0.1100_2 - 0.0100_2 = 0.1000_2$. The result is $0.1000_2 \times 2^3 = 4_{10}$.

## I Do — source worked steps

1. Identify which edges count — Point to CLK: ↑ is LOW → HIGH and samples D. ↓ is HIGH → LOW and does not update Q. Find all three ↑ edges before reading D.

2. ↑1: Q changes from 1 to 0 — CLK rises. Read D = 0. Q was 1, so copying 0 causes a visible change. On the following falling edge Q holds 0, even though CLK moves.

3. D changes, but Q stays 0 — Look where D rises between ↑1 and ↑2. The clock has not risen again, so Q does not follow D. The falling clock edge also leaves Q unchanged.

4. ↑2: Q changes from 0 to 1 — CLK rises again. D is now 1, and Q was 0. Sampling D makes Q change to 1. Q holds at 1 after CLK falls.

5. ↑3: Q stays 1 — despite a rising edge — CLK rises and samples D = 1. But Q already stores 1, so the output does not visibly change. The rule is: at ↑ copy D; otherwise hold Q.

## We Do — guided CHECK

1. D = 0 → Q = 0
Q changes from 1 to 0. Complete row 1.
D = 1 → Q = 1
Q updates from D at ↑2. Complete row 2.
Only a rising edge causes Q to sample D. Complete row 3.
Q becomes 1. Record row 4 and review the complete waveform.

2. Find the update events
At A, B, C, D, E and F, say whether Q changes or holds. Explain using the clock and D input.
A holds; B changes
A: D changes to 1, but CLK remains 0, so Q stays 0. B: CLK rises from 0 to 1, so Q copies D = 1 and changes to 1.
Both hold Q = 1
C: the clock falls (1 → 0), which does not trigger this flip-flop. D: D falls to 0 without a rising clock edge. Q still stores 1.
E changes; F stays 0
E: rising edge samples D = 0, so Q changes from 1 to 0. F: another rising edge samples D = 0, but Q already stores 0 — an update can leave Q unchanged.
Sample on ↑, hold otherwise
A rising clock edge makes Q copy D. Q visibly changes only if the sampled D differs from its stored

value. A falling edge or a change in D alone never updates this rising-edge-triggered flip-flop.

3. Predict Q
Should Q follow D, fall to 0, or stay at 1? Explain your prediction.
Q remains 1
There is no rising clock edge. D changes, but Q retains the previously stored bit.

4. Explain Q
A D-type flip-flop stores one bit because its Q output…
Model explanation
Q stores one bit. At a rising edge Q copies D. Between clock edges, Q retains its stored value.

## You Do — source mark schemes

1. Clocked state
CODE:
Q after triggers:
1, 1, 0, 1
Check: You change Q only at the listed triggers.
Working: at each rising clock edge, copy D to Q. Between clock edges, Q holds the previously stored value.

2. Purpose and inputs
CODE:
A D-type flip-flop stores one bit.
D is the data value to be stored.
CLK controls when D is sampled.
Q is the stored output and holds its value between triggers.
Check: Your answer separates the data input from the timing input.

3. APPLICATION · 3 MARKS
CODE:
A register made from eight D-type flip-flops stores eight bits, one bit per flip-flop. Each D input supplies its next bit; a shared rising clock edge copies the eight input bits to Q together. The Q outputs hold the byte between edges.
Check: Award marks for technically accurate statements tied to the scenario.

4. ANALYSE · 4 MARKS
CODE:
Initially Q = 1. (1) D changes to 0 while CLK is low: Q stays 1. (2) CLK rises with D = 0: Q changes to 0. (3) D changes to 1 while CLK remains high: Q stays 0. Q updates only on the rising clock edge.
Check: Award marks for technically accurate statements tied to the scenario.

5. COMPARE · 3 MARKS
CODE:
An AND gate is combinational: its output depends immediately on its current inputs and it stores no state. A D-type flip-flop stores one bit and Q only updates to D on the active clock edge; Q holds its value between edges.
Check: Award marks for technically accurate statements tied to the scenario.

6. CHALLENGE · 4 MARKS
CODE:

Before the rising edge, the register continues to store 1010 even when D becomes 0111. At the next rising edge, all four flip-flops copy their D inputs and the register stores 0111. Any further D changes without another rising edge leave the stored value at 0111.
Check: Award marks for technically accurate statements tied to the scenario.

## Exit Ticket — source answer

1. True. A D-type flip-flop holds its stored Q value until its triggering clock event.